# Risk Factors for Diabetes Among Adults Aged 40+

## A Combined Logistic Regression and Random Forest Analysis Using NHANES Data

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## Abstract (150-200 words)

## Introduction (500-600 words)

## Methods (600-700 words)

### Data (data sources, justify choice, clean&analyze approach, assumption/limitation)

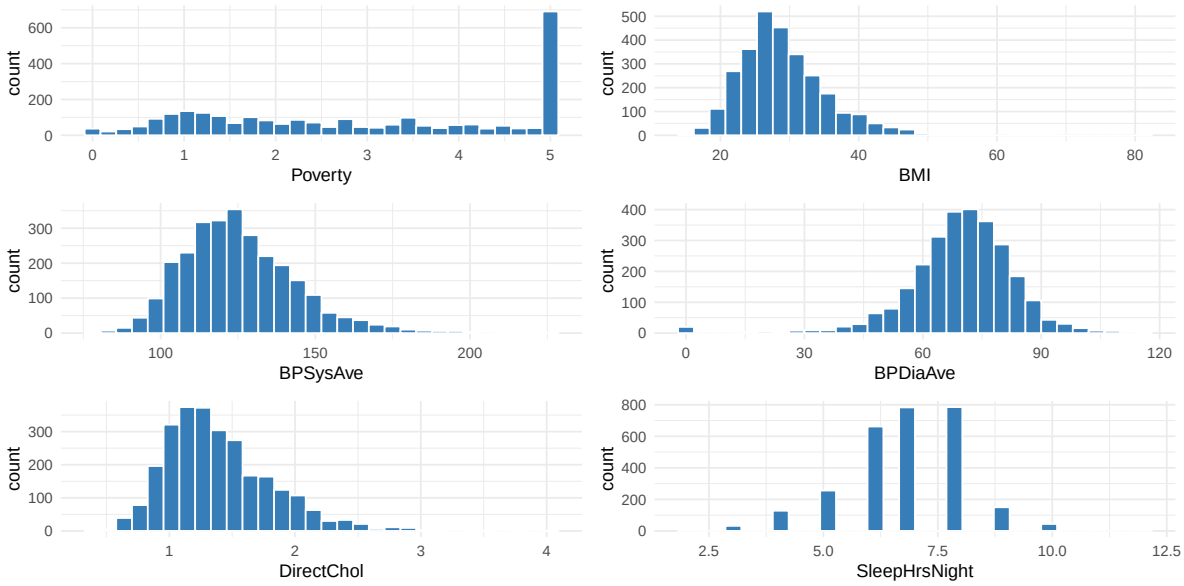



Figure 1: Distribution of Numerical Variables

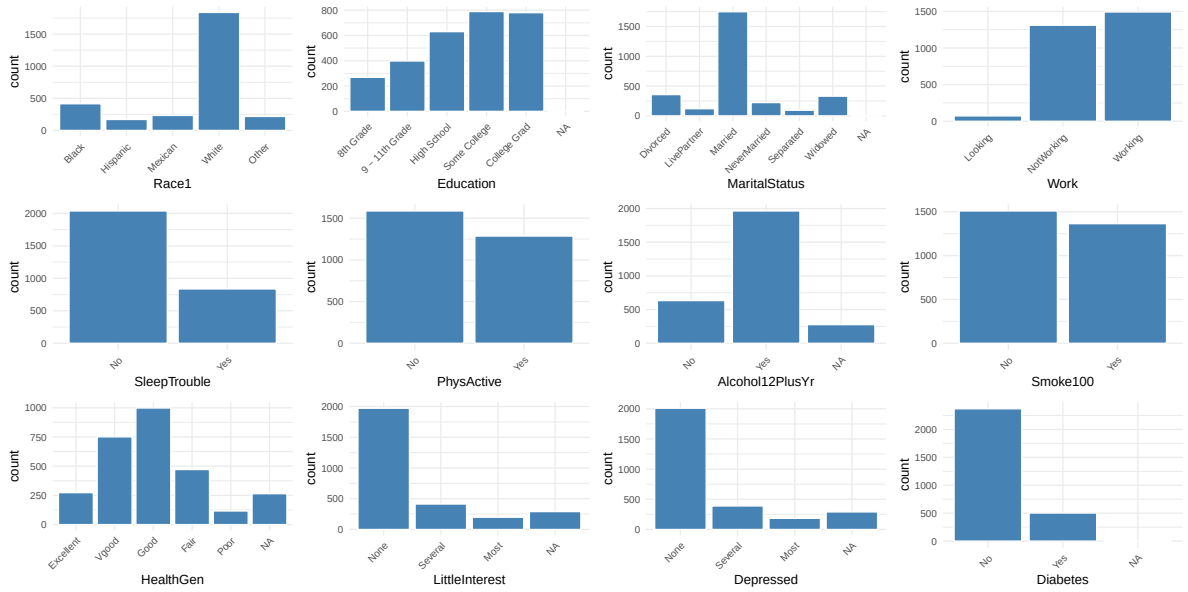



Figure 2: Distribution of Categorical Variables

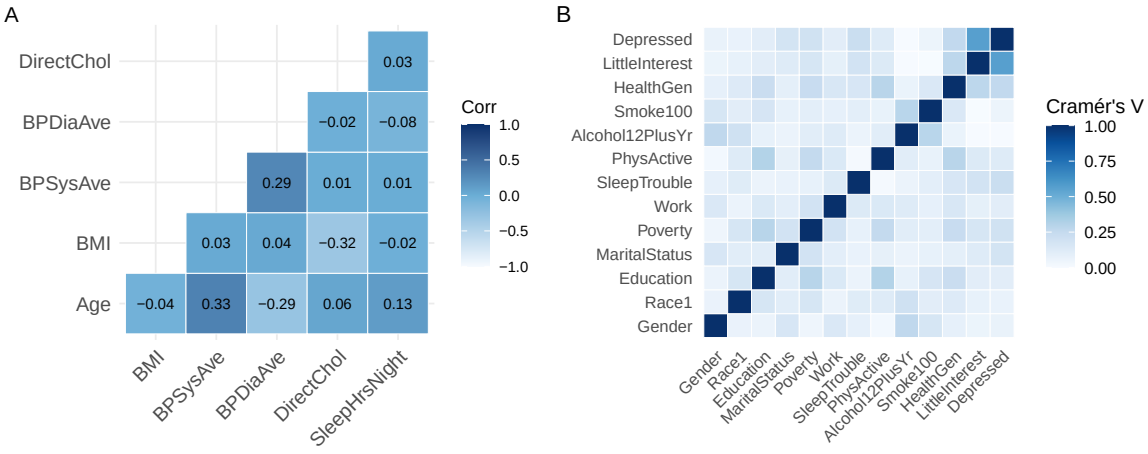



Figure 3: Correlation among Numeric Variables (Left) and Association among Categorical Variables (Right)

# Model (Model selection, model assumption, VIF, model interpretation)

## logistic Regression

To evaluate socioeconomic, behavioral, psychosocial, and clinical determinants of diabetes among adults aged 40 years and older, we fitted a multivariable logistic regression model using the cleaned NHANES dataset. Logistic regression was selected due to the binary nature of the outcome and its interpretability in estimating adjusted odds ratios (ORs) for a range of covariates. This model aligns with our research aim of comparing statistically significant predictors from a parametric model with non-parametric variable importance rankings produced by the Random Forest analysis.

### Model Specification

The full model included one dependent variable and 19 covariates:

**Outcome:** Diabetes
*(Binary indicator of diagnosed diabetes)*

**Control variables**

- **Age** *(Age in years)*
- **Gender** *(Self-reported sex: Male/Female)*
- **Race1** *(Race/Ethnicity category)*

**Socioeconomic factors**

- **Education** *(Highest level of education completed)*
- **Poverty** *(Family poverty-income ratio)*
- **MaritalStatus** *(Current marital status)*
- **Work** *(Employment status)*

**Behavioral factors**

- **BMI** *(Body Mass Index)*
- **BPSysAve** *(Average systolic blood pressure)*

- **BPDiaAve** *(Average diastolic blood pressure)*
- **DirectChol** *(Direct HDL cholesterol measure)*
- **SleepHrsNight** *(Hours of sleep on an average night)*
- **SleepTrouble** *(Trouble sleeping: Yes/No)*
- **PhysActive** *(Physically active in the past 30 days: Yes/No)*
- **Alcohol12PlusYr** *(Had 12 alcoholic drinks in the past year: Yes/No)*
- **Smoke100** *(Smoked at least 100 cigarettes in lifetime: Yes/No)*

**Psychosocial factors**

- **HealthGen** *(Self-rated general health)*
- **LittleInterest** *(Little interest or pleasure in activities: Yes/No)*
- **Depressed** *(Feeling down, depressed, or hopeless: Yes/No)*

The model was specified as:

$$\log\left(\frac{P(D_i=1)}{1-P(D_i=1)}\right) = \beta_0 + \beta_1 X_{\text{gender},i} + \beta_2 X_{\text{age},i} + \beta_3 X_{\text{race1},i} + \beta_4 X_{\text{education},i} + \beta_5 X_{\text{marital},i} + \beta_6 X_{\text{poverty},i}$$
$$+ \beta_7 X_{\text{work},i} + \beta_8 X_{\text{bmi},i} + \beta_9 X_{\text{bpsys},i} + \beta_{10} X_{\text{bpdia},i} + \beta_{11} X_{\text{chol},i} + \beta_{12} X_{\text{sleephrs},i} + \beta_{13} X_{\text{sleeptrouble},i}$$
$$+ \beta_{14} X_{\text{physact},i} + \beta_{15} X_{\text{alcohol},i} + \beta_{16} X_{\text{smoke100},i} + \beta_{17} X_{\text{healthgen},i} + \beta_{18} X_{\text{littleinterest},i} + \beta_{19} X_{\text{depressed},i}$$

for $i = 1, \ldots, 2388$. Continuous predictors were kept on their natural scale to preserve interpretability. Categorical variables were dummy-coded using NHANES reference levels. All predictors were selected based on existing literature and the conceptual framework defined in our research questions.

### Model Estimation

The logistic regression model was estimated using maximum likelihood, a standard approach for binary outcome models. Under this framework, the probability of observing each individual's diabetes status is expressed as a function of the covariates through the logit link, and the overall likelihood is formed by multiplying the Bernoulli probabilities across all observations. The estimation procedure identifies the set of coefficients that maximize this likelihood, producing

parameter estimates that are consistent and allow for valid large-sample inference. The model was fit using the glm() function in R with the binomial family and logit link:

### Multicollinearity Assessment

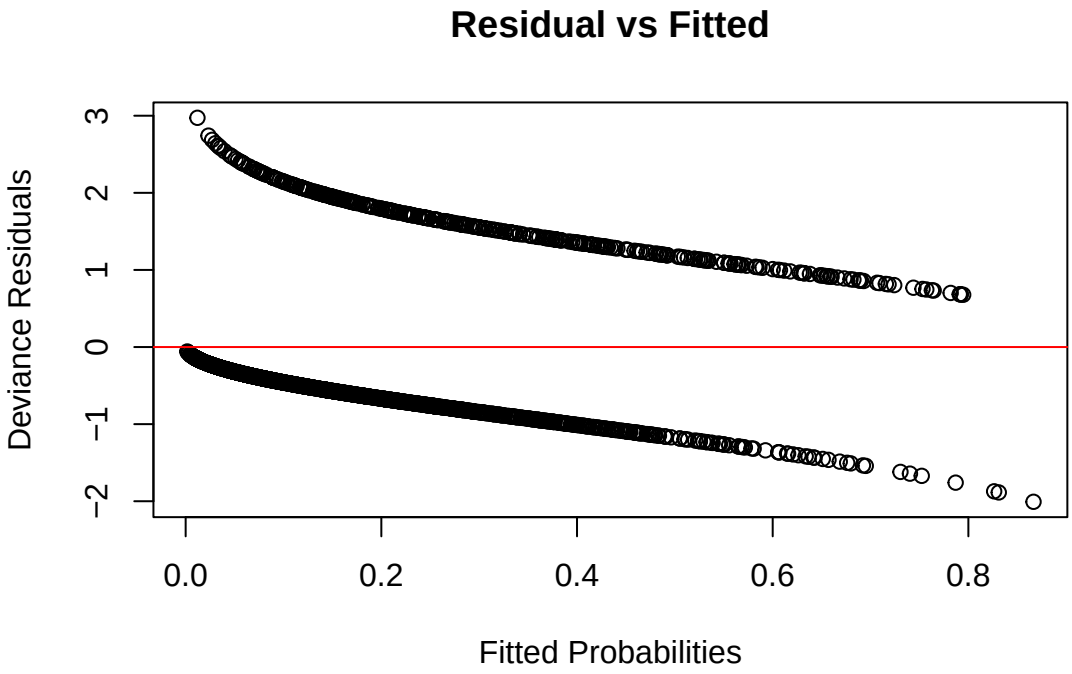
We assessed multicollinearity using the generalized variance inflation factor (GVIF). GVIF reflects how much the variance of a predictor's coefficient is inflated due to correlations with other variables. The "Df" column indicates the predictor's degrees of freedom, which is greater than one for categorical variables with multiple levels. Because GVIF naturally increases with higher Df, we rely on the adjusted statistic $\text{GVIF}^{(1/(2*\text{Df}))}$, which allows comparisons across predictors. Adjusted values close to 1 indicate minimal collinearity, whereas substantially larger values would raise concern. In our model, all adjusted GVIF values were below 2, indicating no meaningful multicollinearity and supporting the stability of the regression estimates.

**Table 1. Generalized Variance Inflation Factors (GVIF) for Predictors in the Logistic Regression Model**

| Variable | GVIF | Df | $\text{GVIF}^{(1/(2*\text{Df}))}$ |
|---|---|---|---|
| Gender | 1.417198 | 1 | 1.190461 |
| Age | 2.067205 | 1 | 1.437778 |
| Race1 | 1.463728 | 4 | 1.048776 |
| Education | 1.393391 | 2 | 1.086471 |
| MaritalStatus | 1.230598 | 2 | 1.053244 |
| Poverty | 1.473221 | 2 | 1.101709 |
| Work | 1.704569 | 2 | 1.142625 |
| BMI | 1.195429 | 1 | 1.093357 |
| BPSysAve | 1.324791 | 1 | 1.150996 |
| BPDiaAve | 1.381903 | 1 | 1.175544 |
| DirectChol | 1.274176 | 1 | 1.128794 |
| SleepHrsNight | 1.167487 | 1 | 1.080503 |
| SleepTrouble | 1.207583 | 1 | 1.098901 |
| PhysActive | 1.203566 | 1 | 1.097072 |
| Alcohol12PlusYr | 1.302351 | 1 | 1.141206 |
| Smoke100 | 1.205641 | 1 | 1.098017 |
| HealthGen | 1.496224 | 4 | 1.051658 |
| LittleInterest | 1.533131 | 1 | 1.238197 |
| Depressed | 1.590547 | 1 | 1.261169 |

### Model Diagnostics

Model assumptions were assessed through inspection of deviance residuals and influential observations. The residual vs. fitted plot showed no systematic pattern, indicating adequate model fit. Cook's distance values were uniformly low, suggesting the absence of influential data points distorting coefficient estimates.



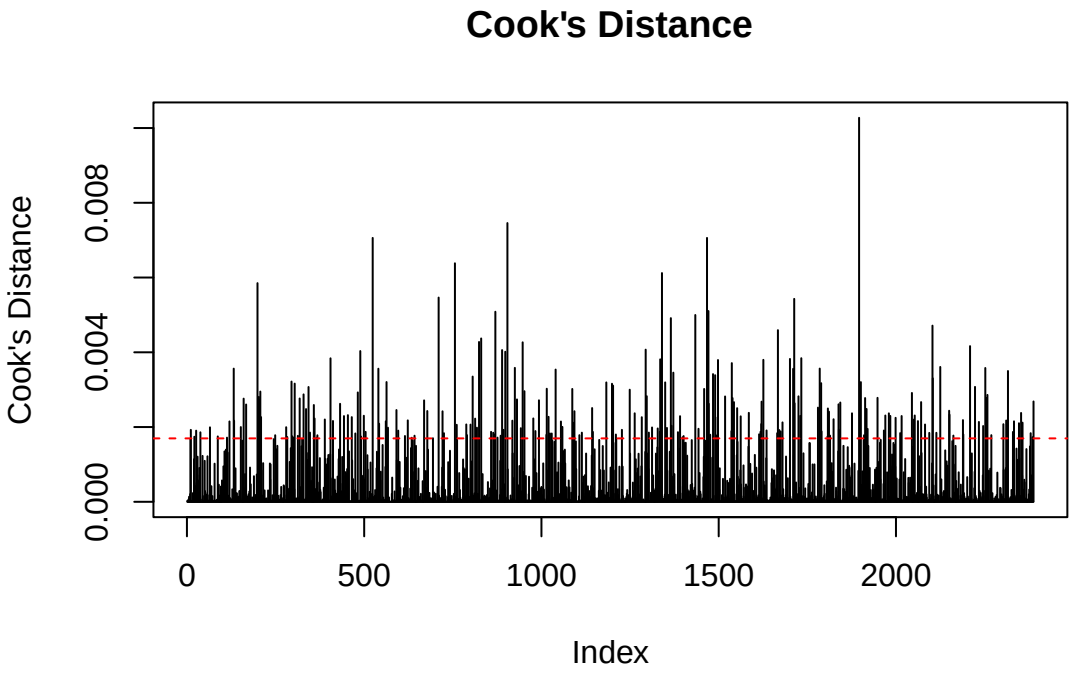




Figure 4: Residual vs Fitted and Cook's Distance Diagnostics

Together, these diagnostics and coefficient patterns support the adequacy of the logistic regression model and provide a strong basis for comparison with the Random Forest results presented later. Because logistic regression assumes linear and additive effects on the log-odds scale, any nonlinear relationships or higher-order interactions may remain undetected. To complement this parametric approach and assess more flexible predictive structures, we next implement a Random Forest model.

# Results (500-600 words)

## Logistic Regression Results

Table 2 summarizes the regression coefficients, standard errors, z-statistics, p-values, and significance indicators. Age, BMI, systolic and diastolic blood pressure, and general health rating emerged as significant predictors of diabetes risk.

**Table 2. Logistic Regression Coefficients for Diabetes Risk**

| Predictor | Estimate | Std. Error | z value | p-value | Significance |
|---|---|---|---|---|---|
| (Intercept) | -3.6113 | 0.9775 | -3.694 | 0.00022 | *** |
| Gender (Male) | 0.0725 | 0.1421 | 0.511 | 0.6096 | |
| Age | 0.0511 | 0.0071 | 7.176 | 7.18e-13 | *** |
| Race: Hispanic | -0.3780 | 0.2833 | -1.334 | 0.1821 | |
| Race: Mexican | -0.1310 | 0.2550 | -0.511 | 0.6095 | |
| Race: White | -0.7934 | 0.1746 | -4.544 | 5.53e-06 | *** |
| Race: Other | 0.1986 | 0.2616 | 0.759 | 0.4479 | |
| Education: Some College | 0.2067 | 0.1473 | 1.404 | 0.1605 | |
| Education: College+ | 0.2193 | 0.1769 | 1.240 | 0.2149 | |
| Married | 0.1654 | 0.2454 | 0.674 | 0.5003 | |
| Not Married | 0.1075 | 0.2536 | 0.424 | 0.6717 | |
| Middle Income | -0.1184 | 0.1786 | -0.662 | 0.5079 | |
| High Income | -0.1149 | 0.2181 | -0.527 | 0.5984 | |
| Work: Not Working | -0.3528 | 0.3858 | -0.915 | 0.3604 | |
| Work: Working | -0.2747 | 0.3840 | -0.715 | 0.4735 | |
| BMI | 0.0598 | 0.0094 | 6.384 | 1.73e-10 | *** |
| Systolic BP | 0.00847 | 0.00364 | 2.323 | 0.0202 | * |
| Diastolic BP | -0.0126 | 0.00543 | -2.322 | 0.0202 | * |
| Direct Cholesterol | -1.0835 | 0.1966 | -5.268 | 1.38e-07 | *** |
| Sleep Hours | -0.0084 | 0.0436 | -0.088 | 0.9296 | |
| Sleep Trouble | 0.2166 | 0.1384 | 1.565 | 0.1175 | |
| Physically Active | 0.1366 | 0.1342 | 1.017 | 0.3089 | |
| Alcohol 12+ / yr | 0.0556 | 0.1521 | 0.365 | 0.7150 | |

| Predictor | Estimate | Std. Error | z value | p-value | Significance |
|---|---|---|---|---|---|
| Ever Smoked 100 Cigarettes | 0.0437 | 0.1309 | 0.334 | 0.7382 | |
| Health: Fair | -0.6661 | 0.2526 | -2.637 | 0.00837 | ** |
| Health: Good | -1.2213 | 0.2539 | -4.810 | 1.51e-06 | *** |
| Health: Very Good | -1.9612 | 0.2892 | -6.782 | 1.19e-11 | *** |
| Health: Excellent | -1.9563 | 0.3513 | -5.570 | 2.55e-08 | *** |
| Little Interest | -0.1791 | 0.1673 | -1.070 | 0.2845 | |
| Depressed | 0.3126 | 0.1718 | 1.820 | 0.0689 | . |

Table 2 summarizes the logistic regression estimates for diabetes among adults aged 40 years and older. Several variables showed statistically significant associations with diabetes status. Age demonstrated a strong positive relationship, with higher age corresponding to higher odds of diabetes. BMI was also significantly associated with increased odds of diabetes. Among the clinical measures, systolic blood pressure was positively associated with diabetes, while diastolic blood pressure showed a small but statistically significant negative association. Direct cholesterol levels were significantly lower among individuals with diabetes.

Self-rated general health exhibited a clear gradient: compared with the reference category of poor health, participants reporting fair, good, very good, or excellent health all showed progressively lower odds of diabetes. These categories displayed some of the largest effect sizes in the model.

Race differences were observed, with White participants showing significantly lower odds of diabetes compared with the Black reference group. Other racial categories did not reach statistical significance.

In contrast, socioeconomic indicators (education, income, marital status, work status) and behavioral factors (physical activity, alcohol use, smoking history, sleep duration, and sleep trouble) were not statistically significant in the logistic regression model.

Overall, the most prominent associations were observed for age, BMI, blood pressure measures, cholesterol, self-rated health, and race, indicating that these variables contributed the most to the adjusted model.

### Model Interpretation

In this framework, each coefficient represents the change in the log-odds of having diabetes associated with a one-unit increase in a continuous predictor or a change from the reference category for categorical predictors, holding all other variables constant. Exponentiating the coefficients yields adjusted odds ratios, which provide a more interpretable scale for assessing relative increases or decreases in diabetes risk. An odds ratio greater than 1 indicates

an elevated likelihood of diabetes, whereas an odds ratio below 1 suggests a protective association. The intercept $\beta_0$ represents the baseline log-odds of diabetes for an individual in all reference categories with continuous predictors set to zero, serving as the model's starting point for estimating individual-level risk. This interpretation allows us to compare the relative contributions of demographic, behavioral, psychosocial, and clinical factors within a unified statistical model and provides a basis for evaluating how these associations align with the variable importance patterns observed in the Random Forest analysis.

To further assess variable importance and capture potential nonlinear relationships or interactions, a Random Forest model was subsequently applied using the same set of predictors.

## Discussion (600-700 words)

conclusion, limitation

## References